

Clinical Features of Raynaud Phenomenon

During attacks, one or more fingers or toes may become numb and feel "dead." Upon re-warming, the affected digits often turn bright red and may be accompanied by throbbing pain. When pain is prominent during the ischemic phase, a secondary cause should be suspected. Episodes typically last from minutes to hours. The fingers and toes are most commonly affected, though the nose, earlobes, and nipples can also be involved.

A thorough review of systems is essential to evaluate for underlying conditions. Screen for symptoms suggestive of connective tissue disease (e.g., arthralgias, arthritis, dysphagia, heartburn, rash, photosensitivity, skin changes, muscle weakness, shortness of breath, sicca symptoms), drug-induced causes, signs of obstructive arterial disease (such as intermittent claudication), and exposure to vibratory tools or repetitive finger trauma.

Cutaneous Lesions

Stress-induced attacks of Raynaud phenomenon are often observed during the patient interview or physical examination. Characteristic findings include well-demarcated blanching or cyanosis extending from the tip of the digit proximally. The skin distal to the ischemic line is cold, while the proximal skin remains pink and warm. With re-warming, blanched areas may first become cyanotic due to sluggish blood flow, then turn bright red as reactive hyperemia occurs. Persistent ischemic discoloration suggests a secondary cause.

Careful inspection of the digits should assess for trophic or ischemic changes, which indicate chronic or severe disease. Skin may appear atrophic, thin, and tight—termed **sclerodactyly**—with loss of dorsal hair. Nails may become brittle and malformed. Painful ulcerations can develop on the finger pads or around the nail folds, often worsening at night. These ulcers heal slowly, may

become infected, and typically leave small, pitted scars. Digital gangrene is rare but possible in advanced cases.

Related Physical Findings

The physical examination should include assessment of all peripheral pulses and bilateral arm blood pressure measurements. **Allen's test** is useful for evaluating arterial inflow to the hands; abnormal results suggest structural microvascular disease and raise concern for secondary Raynaud phenomenon.

To perform Allen's test: - Compress both the radial and ulnar arteries while the patient opens and closes the fist until the palm blanches. - Release pressure on one artery at a time and observe the speed of color return, indicating adequacy of arterial supply.

Assess for vascular obstruction due to **thoracic outlet syndrome** using the **Adson maneuver**: evaluate for diminished radial pulse with neck extension and rotation or arm abduction.

A detailed neurologic examination should be performed to detect signs of sympathetic hyperactivity, abnormal reflexes, muscle weakness or atrophy, or **carpal tunnel syndrome** (median nerve compression).

Skin examination may reveal **telangiectases**, abnormal skin texture, rashes, or **purpura**, which can provide clues to underlying connective tissue disease or hyperviscosity syndromes.

Nail fold capillary microscopy may show **capillary dropout**, along with enlarged, dilated, and tortuous capillary loops—commonly seen in scleroderma (Fig. 171-3).